Supplementary Material

Attention-Based Safe Reinforcement Learning with a Feasibility-Preserving Safety Layer for Size-Agnostic V2G Charging Control

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This document supplements the main text. Section S1 proves the formal results, Section S2 documents the modelling in EV2Gym, and Section S3 gives the request pipeline and the perfect-information optima. The compared methods, the training, the tuning and the statistics follow in Section S4, and Section S5 collects further results. Equations, lemmas and figures of the main text are cited with their main-text numbers, and items of this document carry the prefix S. Abbreviations follow the main text on electric vehicle (EV) charging with vehicle-to-grid (V2G) services and photovoltaic (PV) generation at a charge point operator (CPO) station. The method is the safe attention-based controller (SA-Attn), trained with proximal policy optimization (PPO) and compared with a multilayer perceptron (MLP), economic model predictive control (eMPC) and a heterogeneous graph policy (EV-GNN). Further abbreviations are the linear program (LP), the Karush–Kuhn–Tucker (KKT) conditions, least-laxity-first (LLF) scheduling and conformal prediction (CP). Table S1 lists the main symbols.

S1 Proofs

This section proves the formal results of the main text and states the auxiliary lemmas that the main text cites. All statements hold for every EV separately, and the index i is dropped where no confusion arises. Inequalities are exact in real arithmetic, and the implementation checks them with one tolerance of 10^{-9} in kW or kWh that the unused part of the margin ε_q absorbs.

S1.1 Size independence of the attention policy

Lemma S1 (Equivariance and size independence)**.** *Let ϖ permute the charger tokens and their masks while the global token stays fixed.*

(i) The means μ_θ and standard deviations σ_θ are permutation-equivariant, and the value V_ω is permutation-invariant.

(ii) Outputs of unmasked tokens and the value do not depend on the features or the number of masked tokens.

(iii) The number of parameters does not depend on N.

Proof. Embeddings, layer normalisation, feed-forward blocks and the actor head act row-wise, so they commute with the permutation. For one attention head, the key mask depends only on the key index, so permuting tokens and mask maps the mask matrix M to $PMP^\top$ for the permutation matrix P that fixes the global token. Row-wise softmax satisfies $\mathrm{softmax}(PXP^\top) = P\,\mathrm{softmax}(X)P^\top$, so a masked attention head maps permuted inputs to permuted outputs, and composition gives (i) for the actor. A masked key receives zero weight in every softmax, so its value never enters the output of another token, and together with the row-wise structure this property gives (ii). Attention pooling sums over keys and is therefore invariant, and every weight matrix has a shape fixed by $(F_c, F_g, d_m, n_H, n_L)$ alone, so (iii) holds. The safety layer is permutation-equivariant as well, because its problem is invariant under joint permutation of p, x, l and u and its minimiser is unique. □

Table S1: Main symbols. Units are given in brackets; quantities of EV i carry the index of its charger.

Symbol	Meaning	Symbol	Meaning
N, $\mathcal{N}(t)$, $n(t)$	chargers, occupied chargers, connected EVs	Δt, T	step length [h], horizon [steps]
t_i^{arr}, d_i	arrival step, last controlled step	$\tau_i(t)$	remaining actions, $d_i - t + 1$
$E_i(t)$, E_i^{cap}, $E_i^{\min}$	battery energy, capacity, minimum [kWh]	E_i^{des}, E_i^{req}	desired and admitted energy [kWh]
p_i, $\tilde{p}_i$	commanded and reported power [kW]	g_i, η_i^{c}, η_i^{d}	energy map and efficiencies
$\bar{P}_i(E)$, $\bar{P}_i^{\mathrm{nom}}$	maximum charging power, its nominal value [kW]	$p_i^{\min}(E)$, P_i^{dis}	physical lower limit, discharge power [kW]
P_{tr}, ρ_{tr}	transformer rating, rating per charger [kW]	P^{inf}, P^{pv}	inflexible load, PV generation [kW]
C, C_{rev}, $\tilde{C}$	forward, reverse and tightened forward limit [kW]	ε_q	actuation margin, 0.008 kW
c^{ch}, c^{dis}, c^{thr}	charging, discharge and throughput price [€/kWh]	$\varepsilon_{\mathrm{sat}}$	saturation threshold, 0.001 kW
φ_i, $\Psi_{i,k}$, R_i	one-step full-power map, k-fold map, reachable energy	$B_i(k)$, $\hat{B}_i(k)$	exact and certified thresholds [kWh]
l_i, u_i	lower and upper bound of the safety layer [kW]	χ_i	laxity [steps]
x_i, a_i, P^{ref}	raw action [kW], raw sample, scale 11 kW	$\mathcal{K}(t)$, Π_t	feasible set, projection
λ^*, $\mathcal{F}$	common shift [kW], free set	b	fallback budget [kW]
ς, ς^{tr}, m, κ	state of charge, taper start, multiplier, exponent	$z_0(m)$, $\kappa^*(m)$	exact-delivery constants
ϱ_i^{in}, ϱ_i^{out}	inner and outer taper slope [1/h]	$G_i^{\max}$	rounding allowance [kWh]
J^π, J^{pen}	profit, penalised profit [€]	$J^{\mathrm{pen,LP}}$, $J^{\mathrm{LP,ceil}}$	soft and hard optimum [€]
w_E, w_P, w	evaluation and training penalty weights [€/kWh]	ΔE_i^{sh}	departure shortfall [kWh]
ξ_i, ξ_0	charger and global token	d_m, n_L, n_H	width, layers, heads
θ, ω	actor and critic parameters	β	regulariser weight
γ, λ_{GAE}, ϵ_{clip}	discount, GAE parameter, clip range	n_{mb}, c_{vf}, c_{ent}	minibatch size, loss coefficients
Λ_E, Λ_P, η_Λ	Lagrange multipliers, dual step size	α, n_{cal}	miscoverage level, calibration size
$\hat{q}$, e_j, $\hat{Q}_{1-\alpha}$	quantile predictor, score, conformal quantile	$\hat{d}_i^{\mathrm{lo}}$, τ_i^{lo}	departure bound, remaining steps

The parameter counts are 610,370 for the actor and 676,481 for the critic, 1,286,851 in total for every N. For the small ablation network ($d_m = 64$, $n_L = 2$, feed-forward width 256, head width 32), actor and critic have 228,067 parameters in total. Deep Sets applies a per-token network to each charger token together with the masked mean of all token embeddings and the global token, so the same argument covers it. Its width matches the parameter count of SA-Attn within 10%. The actor of the MLP baseline with 20 slots and two hidden layers of 640 units has 614,440 parameters, a number that grows linearly with the number of slots.

S1.2 Reachable energy and thresholds

Lemma S2 (Reachable energy)**.** *Under A1, Ψ_k is continuous and non-decreasing for every $k \geq 0$. Any admissible power sequence from $E(t)$ gives $E(t+k) \leq \Psi_k(E(t))$, with equality for full-power charging. An EV is individually feasible at step t, meaning that some admissible control meets Equation 4, if and only if $E(t) \geq B(\tau(t))$.*

Proof. A composition of continuous non-decreasing maps is continuous and non-decreasing, so $\Psi_k = \varphi^k$ inherits both properties from A1. For the reach bound we use induction on k, where the case $k = 0$ is trivial. In the induction step, an admissible power $p \leq \bar{P}(E)$ gives $E(t+1) = E + \Delta t\, g(p) \leq E + \Delta t\, g(\bar{P}(E)) = \varphi(E)$ because g is increasing, and the induction hypothesis from $E(t+1)$ with the monotonicity of Ψ_{k-1} gives

$$E(t+k) \leq \Psi_{k-1}\big(E(t+1)\big) \leq \Psi_{k-1}\big(\varphi(E)\big) = \Psi_k(E).$$

Full-power charging gives equality at every step.

For the threshold characterisation we show $\{E \in [E^{\min}, E^{\mathrm{cap}}] : \Psi_k(E) \geq E^{\mathrm{req}}\} = [B(k), E^{\mathrm{cap}}]$ by induction, where the case $k = 0$ is the definition $B(0) = E^{\mathrm{req}}$. The set in Equation 9 is non-empty

because $\varphi(E^{\mathrm{cap}}) = E^{\mathrm{cap}} \geq E^{\mathrm{req}} \geq B(k-1)$. In the induction step, $\Psi_k(E) = \Psi_{k-1}(\varphi(E)) \geq E^{\mathrm{req}}$ holds if and only if $\varphi(E) \geq B(k-1)$, and since φ is continuous and non-decreasing, $\{E : \varphi(E) \geq B(k-1)\}$ is a closed up-set with minimum $B(k)$. By the reach bound, some admissible control meets Equation 4 from $E(t)$ if and only if full-power charging does, that is, if and only if $\Psi_{\tau(t)}(E(t)) \geq E^{\mathrm{req}}$. □

For a certified table, the first two properties of Equation 11 imply $\hat{B} \geq B$ componentwise, because $\varphi(\hat{B}(k)) \geq \hat{B}(k-1) \geq B(k-1)$ gives $\hat{B}(k) \geq B(k)$ by the minimality in Equation 9.

S1.3 Projection

Proof of Proposition 1. (i) If $p \in \mathcal{K}(t)$, then $l \leq p \leq u$ gives $l \leq u$ and $\sum_i l_i \leq \mathbf{1}^\top p \leq \sum_i u_i$, and combined with $-C_{\mathrm{rev}} \leq \mathbf{1}^\top p \leq \tilde{C}$ these inequalities give the four conditions. Conversely, $p(\upsilon) = l + \upsilon(u - l)$ stays in the box for $\upsilon \in [0, 1]$, and $\mathbf{1}^\top p(\upsilon)$ takes every value in $[\sum_i l_i, \sum_i u_i]$, an interval that meets $[-C_{\mathrm{rev}}, \tilde{C}]$ under the four conditions.

(ii) The problem in Equation 12 minimises a strictly convex quadratic over a non-empty polyhedron, so its minimiser exists and is unique, and the KKT conditions are necessary and sufficient. With multipliers $\mu^+, \mu^- \geq 0$ for the aggregate constraints and $\nu^l, \nu^u \geq 0$ for the box, they read

$$p - x + (\mu^+ - \mu^-)\mathbf{1} - \nu^l + \nu^u = 0, \qquad \mu^+(\mathbf{1}^\top p - \tilde{C}) = 0, \qquad \mu^-(\mathbf{1}^\top p + C_{\mathrm{rev}}) = 0,$$
$$\nu_i^l(p_i - l_i) = 0, \qquad \nu_i^u(u_i - p_i) = 0.$$

The aggregate response $\mathcal{S}(\lambda) = \sum_i \mathrm{clip}(x_i - \lambda, l_i, u_i)$ is continuous, non-increasing and piecewise linear, with $\mathcal{S}(\lambda) = \sum_i u_i$ for $\lambda \leq \min_i(x_i - u_i)$ and $\mathcal{S}(\lambda) = \sum_i l_i$ for $\lambda \geq \max_i(x_i - l_i)$. If $\mathcal{S}(0) > \tilde{C} \geq \sum_i l_i$, the intermediate value theorem gives a root of $\mathcal{S}(\lambda) = \tilde{C}$, and every such root is positive; the case $\mathcal{S}(0) < -C_{\mathrm{rev}} \leq \sum_i u_i$ is symmetric with a negative root. Now set $p = \mathrm{clip}(x - \lambda^*, l, u)$, $\mu^+ = \max(\lambda^*, 0)$ and $\mu^- = \max(-\lambda^*, 0)$, and choose the box multipliers coordinate by coordinate. Free coordinates with $l_i < x_i - \lambda^* < u_i$ receive $\nu_i^l = \nu_i^u = 0$, coordinates with $x_i - \lambda^* \leq l_i$ receive $\nu_i^l = l_i - x_i + \lambda^* \geq 0$ and $\nu_i^u = 0$, and the remaining ones receive $\nu_i^u = x_i - \lambda^* - u_i \geq 0$ and $\nu_i^l = 0$. Stationarity then holds coordinate-wise, and complementarity holds because $\lambda^* > 0$ only when $\mathbf{1}^\top p = \tilde{C}$ and $\lambda^* < 0$ only when $\mathbf{1}^\top p = -C_{\mathrm{rev}}$. For $\lambda^* = 0$ the point with $\mathbf{1}^\top p = \mathcal{S}(0)$ is feasible, so in all cases $p = p^*$. If $\lambda_1 < \lambda_2$ are both roots, every coordinate $\mathrm{clip}(x_i - \lambda, l_i, u_i)$ is non-increasing in λ and the sums agree, so all coordinates agree. □

PPO treats the safety layer as part of the environment and never uses its Jacobian. The Jacobian nevertheless describes how the layer passes changes of the raw action to the commanded powers.

Proposition S1 (Jacobian of the projection)**.** *Let $\mathcal{F} = \{i : l_i < x_i - \lambda^* < u_i\}$ be the free set and D the diagonal matrix of its indicators. Wherever $p^* = \Pi_t(x)$ is differentiable, its Jacobian is D if the aggregate constraint is inactive and $D - \mathbf{1}_{\mathcal{F}}\mathbf{1}_{\mathcal{F}}^\top/|\mathcal{F}|$ if it is active with $\mathcal{F} \neq \emptyset$, where $\mathbf{1}_{\mathcal{F}}$ is the indicator vector of $\mathcal{F}$.*

Proof. If the aggregate constraint is inactive, locally $p_i = x_i$ for $i \in \mathcal{F}$ and $p_i \in \{l_i, u_i\}$ is constant otherwise, so the Jacobian is D. With the aggregate constraint active at the target $\vartheta \in \{\tilde{C}, -C_{\mathrm{rev}}\}$, instead, $\sum_{i \in \mathcal{F}}(x_i - \lambda) + \sum_{i \notin \mathcal{F}} p_i = \vartheta$. Hence $\partial\lambda/\partial x_j = \mathbb{1}[j \in \mathcal{F}]/|\mathcal{F}|$ and $\partial p_i/\partial x_j = \mathbb{1}[i \in \mathcal{F}]\big(\mathbb{1}[i = j] - \mathbb{1}[j \in \mathcal{F}]/|\mathcal{F}|\big)$. If $\mathcal{F} = \emptyset$, the point p is locally constant, and on the null set where p^* is not differentiable, the matrices of the adjacent regions are elements of Clarke's generalised Jacobian. □

The Jacobian is symmetric and idempotent, and as a Euclidean projection onto a convex set, Π_t is non-expansive. On an active row the Jacobian is zero for all clipped coordinates, the zero-gradient effect that motivates treating the layer as part of the environment [1].

S1.4 One-step invariance

The proof sketch of Theorem 1 in the main text and its full proof below both rest on the following lemma.

Lemma S3 (One-step invariance)**.** *Assume A1 and A3, and let $\hat{B}$ satisfy Equation 11. Let EV i be connected at step t with $\tau = \tau_i(t)$ remaining actions and energy $E = E_i(t) \geq \hat{B}(\tau)$. Then $p^{\min}(E) \leq l_i(t) \leq u_i(t)$, and every commanded power $p \in [l_i(t), u_i(t)]$ leads to $E_i(t+1) \geq \hat{B}(\tau - 1)$.*

Proof. The bound $p^{\min} \leq l_i$ holds because $p^{\min} \leq 0 \leq u_i$ and both arguments of the outer minimum in Equation 10 are at least $p^{\min}$. For $l_i \leq u_i$ the outer minimum suffices, and the guard is inactive in exact arithmetic, because $g^{-1}\big((\hat{B}(\tau-1) - E)/\Delta t\big) \leq \bar{P}(E)$ is equivalent to $\hat{B}(\tau - 1) \leq \varphi(E)$, and A1 with Equation 11 gives $\varphi(E) \geq \varphi(\hat{B}(\tau)) \geq \hat{B}(\tau - 1)$. Turning to the second claim, let $y = (\hat{B}(\tau-1) - E)/\Delta t$, and since $l_i \geq \min\{u_i, g^{-1}(y)\} = g^{-1}(y)$ in exact arithmetic and g is increasing, every $p \geq l_i$ satisfies $g(p) \geq y$, and A3 gives $E_i(t+1) \geq E + \Delta t\, g(p) \geq \hat{B}(\tau - 1)$. □

When $\tau \geq 2$, the EV is individually feasible at $t+1$ because $\tau_i(t+1) = \tau - 1$, and when $\tau = 1$, the lemma gives $E_i(d_i + 1) \geq \hat{B}(0) \geq E_i^{\text{req}}$.

S1.5 Feasibility preservation

The proof of Theorem 1 treats feasible steps, infeasible steps and the departure part in turn. Claim (a) uses only A3 and A6, and the departure part uses A1, A2 and A4 in addition. The conditions A1–A6 are those listed in Section 4.4, with A5 the definition of a feasible step.

Proof of Theorem 1. Claim (a) on feasible steps. The guard in Equation 10 gives $p_i^{\min} \leq l_i \leq u_i$ for every connected EV at every step, whether or not the EV is individually feasible, and empty chargers have $l_i = u_i = 0$. If the feasibility test of Algorithm S1 passes, Proposition 1(i) gives $\mathcal{K}(t) \neq \emptyset$, so $p(t) = \Pi_t(x(t)) \in \mathcal{K}(t)$ for every finite $x(t)$. The physical limits follow from $p_i(t) \in [l_i, u_i] \subseteq [p_i^{\min}(E_i(t)), \bar{P}_i(E_i(t))]$. For the transformer limit, $\mathbf{1}^\top p(t) \in [-C_{\text{rev}}(t), \tilde{C}(t)]$, and A3 gives

$$\sum_i \tilde{p}_i(t) \in \Big[\sum_i p_i(t),\ \sum_i p_i(t) + n(t)\varepsilon_q\Big] \subseteq \big[-C_{\text{rev}}(t),\ \tilde{C}(t) + n(t)\varepsilon_q\big] = \big[-C_{\text{rev}}(t), C(t)\big].$$

Claim (a) on infeasible steps. Under A6, $\tilde{C}(t) = C(t) - n(t)\varepsilon_q \geq (N - n(t))\varepsilon_q \geq 0$ and $C_{\text{rev}}(t) \geq 0$, so case (i) of the fallback rule needs $\tilde{C} < -C_{\text{rev}} \leq 0$ and is therefore impossible. Since $u_i \geq 0$, case (ii) with $\sum_i u_i < -C_{\text{rev}} \leq 0$ is impossible as well. Sub-case (iv) needs $b = \tilde{C} - \sum_i p_i^{\min} < 0$ and cannot occur because $p_i^{\min} \leq 0$. Every infeasible step is therefore a step of case (iii), and Lemma S4 gives $p^{\min} \leq p \leq l \leq u$ and $\mathbf{1}^\top p = \tilde{C} \in [-C_{\text{rev}}, \tilde{C}]$. The physical limits follow, and A3 transfers the sum bound to the realised powers exactly as on feasible steps.

Claim (b). Fix an EV i with $p_i(t) \geq l_i(t)$ at every step of its stay, and consider the invariant $E_i(t) \geq \hat{B}_i(\tau_i(t))$. At arrival the invariant is A4. If it holds at step t, then $l_i(t)$ follows the first case of Equation 10, and every output of the projection or the fallback rule satisfies $p_i(t) \leq u_i(t)$, so $p_i(t) \in [l_i(t), u_i(t)]$. Lemma S3 then gives $E_i(t+1) \geq \hat{B}_i(\tau_i(t) - 1)$. For $d_i > t$ this inequality is the invariant at $t+1$, and for $d_i = t$ it is the departure requirement $E_i(d_i + 1) \geq \hat{B}_i(0) \geq E_i^{\text{req}}$. The argument uses only the commands sent to EV i, so it holds whatever happens at the other chargers. Any EV whose stay contains no infeasible step receives $p_i \in [l_i, u_i]$ at every step of its stay and is covered. Since EV2Gym removes every EV by step $T - 2$, every EV of the episode reaches its departure inside the horizon, and the departure part of the argument applies to all of them.

Claim (c). If A5 holds at every step, no step of the episode is infeasible, and claim (b) therefore applies to every EV of the episode. □

A2 enters through the remaining actions $\tau_i(t)$ that the bounds need at every step. Admission control never changes a requirement after arrival, so the requirement used in the invariant is fixed during the stay.

A policy-independent sufficient condition for A5. The lower bound l_i is non-increasing in E, because both terms of its first case are, and we let $l_i^{\max}(t)$ denote l_i evaluated at $E = \hat{B}_i(\tau_i(t))$. Suppose that, at every step, $C_{\text{rev}}(t) \geq 0$, $\tilde{C}(t) \geq -C_{\text{rev}}(t)$ and $\sum_{i \in \mathcal{N}(t)} l_i^{\max}(t) \leq \tilde{C}(t)$. Under the invariant, $l_i(t) \leq l_i^{\max}(t)$ and $\sum_i u_i \geq 0 \geq -C_{\text{rev}}$, so A5 holds at t, and by induction A5 then holds for every policy. However, $l_i^{\max}(t) = \bar{P}_i(\hat{B}_i(\tau_i))$ whenever $\hat{B}_i(\tau_i) > E_i^{\min}$, so the condition asks the transformer to serve all non-trivially constrained EVs at full power at the same time.

S1.6 Fallback rule and deficit bound

When the feasibility test fails, the fallback rule replaces the projection. Since $l \le u$ holds by construction, three cases remain, the third with a sub-case.

(i) Contradictory limits, $\tilde{C} < -C_{\mathrm{rev}}$: the procedure of case (iii) is applied, so the forward limit has priority.

(ii) Excess PV export, $\sum_i u_i < -C_{\mathrm{rev}}$: every occupied charger receives $p_i = u_i$.

(iii) Excess required power, $\sum_i l_i > \tilde{C}$: every EV starts at $p_i^{\min}$, and the budget $b = \tilde{C} - \sum_i p_i^{\min}$ is distributed in increasing order of laxity, each EV receiving up to $l_i - p_i^{\min}$.

(iv) Sub-case of (i) and (iii) with $b < 0$: the output stays at $p = p^{\min}$, the closest admissible command to the forward limit.

In case (iii), ties are broken by the larger gap $l_i - p_i^{\min}$ and then by the lower charger index. The logs label the cases (i), (ii), (iii) and (iv) as FB-1*, FB-2, FB-1 and FB-0 with the numeric codes 3, 2, 1 and 0, and every analysis table uses this mapping. Under A6, cases (i), (ii) and (iv) cannot occur (proof of Theorem 1), so their counts in the result tables verify A6 on every replay.

Lemma S4 (Fallback)**.** *In case (iii) of the fallback rule with $b \ge 0$, the output satisfies $p^{\min} \le p \le l$ and $\mathbf{1}^\top p = \tilde{C}$, so the transformer limit and the physical limits hold. Every EV that is individually feasible at step t and receives $p_i = l_i$ stays individually feasible. In cases (ii) and (iv), the output minimises the transformer violation over the box $[p^{\min}, u]$, so no admissible command avoids that violation.*

Proof. In case (iii), $\sum_i (l_i - p_i^{\min}) > \tilde{C} - \sum_i p_i^{\min} = b$, so the budget is exhausted before every EV reaches l_i; hence $\mathbf{1}^\top p = \tilde{C}$ and $p \le l$. Outside case (i) we also have $\tilde{C} \ge -C_{\mathrm{rev}}$, and an individually feasible EV with $p_i = l_i$ stays feasible by Lemma S3. Over the box $[p^{\min}, u]$, the sum $\mathbf{1}^\top p$ is smallest at $p^{\min}$ and largest at u, so the outputs of cases (iv) and (ii) minimise the commanded violation. □

Lemma S5 (Deficit bound)**.** *Assume A1, A3 and A4, and assume that $\bar{P}_i$ is non-increasing in E. With $\eta^{\mathrm{c}} = \eta^{\mathrm{d}} = 1$ as in our EV2Gym instantiation, the total departure shortfall of the fleet in an episode is at most*

$$\Delta t \sum_{t \in \mathcal{T}_{\mathrm{iii}}} \Big(\sum_i l_i(t) - \tilde{C}(t) \Big) + \Delta t \sum_{t \in \mathcal{T}_{\mathrm{iv}}} \sum_i \big(l_i(t) - p_i^{\min}(t) \big), \tag{S1}$$

where $\mathcal{T}_{\mathrm{iii}}$ and $\mathcal{T}_{\mathrm{iv}}$ are the steps of cases (iii) and (iv).

Lemma S6 in Section S2 shows that EV2Gym's taper satisfies the extra condition of Lemma S5.

Proof. Let $\tilde{l}_i(t) = l_i(t)$ if $E_i(t) \ge \hat{B}_i(\tau_i(t))$ and $\tilde{l}_i(t) = u_i(t)$ otherwise, and define the feasibility deficit $\Delta E^{\mathrm{def}}(t) = (\hat{B}(\tau(t)) - E(t))^+$ with $\tau(d_i{+}1) = 0$. The shortfall $(E^{\mathrm{req}} - E(d_i{+}1))^+$ is at most $\Delta E^{\mathrm{def}}(d_i{+}1)$ because $\hat{B}(0) \ge E^{\mathrm{req}}$, and $\Delta E^{\mathrm{def}}(t_i^{\mathrm{arr}}) = 0$ holds by A4. It therefore suffices to show $\Delta E^{\mathrm{def}}(t+1) \le \Delta E^{\mathrm{def}}(t) + \Delta t \left(g(\tilde{l}) - g(p) \right)^+$ at every step and to sum these increments over the stay. If $E \ge \hat{B}(\tau)$, then $E + \Delta t\, g(l) \ge \hat{B}(\tau - 1)$ by Equation 10, and A3 gives $E(t+1) \ge \hat{B}(\tau - 1) - \Delta t\, (g(l) - g(p))^+$. When $E < \hat{B}(\tau)$ instead, $\tilde{l} = u$ and $E + \Delta t\, g(u) = \varphi(E)$, and Equation 11 with the 1-Lipschitz property of φ gives $\hat{B}(\tau - 1) - \varphi(E) \le \varphi(\hat{B}(\tau)) - \varphi(E) \le \hat{B}(\tau) - E = \Delta E^{\mathrm{def}}(t)$. Hence $\hat{B}(\tau - 1) - E(t+1) \le \Delta E^{\mathrm{def}}(t) + \Delta t\, (g(u) - g(p))$ in this case as well. For the fleet bound with g the identity, $p_i \ge \tilde{l}_i$ holds on feasible steps and $p = u$ holds in case (ii) of the fallback rule. In case (iii), $p \le l$ and $\sum_i (l_i - p_i) = \sum_i l_i - \tilde{C}$, and in case (iv), $p = p^{\min}$, so summing the per-step increments over all EVs gives Equation S1. □

S1.7 Conformal departure guarantee

Proof of Proposition 2. (i) By exchangeability, the rank of the test score $e_{n_{\mathrm{cal}}+1} = \hat{q}(\mathbf{w}_i) - d_i$ among the scores $e_1, \ldots, e_{n_{\mathrm{cal}}+1}$ is uniform up to ties, so $\mathbb{P}(e_{n_{\mathrm{cal}}+1} \le \hat{Q}_{1-\alpha}) \ge \lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / (n_{\mathrm{cal}}+1) \ge 1 - \alpha$ [2,3]. The event $e_{n_{\mathrm{cal}}+1} \le \hat{Q}_{1-\alpha}$ is equivalent to $d_i \ge \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha}$, and because d_i is an integer, also to $d_i \ge \lceil \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha} \rceil$. Taking the maximum with t_i^{arr} changes nothing, because $d_i \ge t_i^{\mathrm{arr}}$ holds surely.

(ii) Consider the event that $d_i \geq \hat{d}_i^{\mathrm{lo}}$ and that $p_i(t) \geq l_i(t)$ at every step of the stay. On this event, Lemma S3 with τ^{lo}, applied up to $t = \hat{d}_i^{\mathrm{lo}}$, gives $E_i(\hat{d}_i^{\mathrm{lo}} + 1) \geq \hat{B}(0)$. For the remaining steps $\hat{d}_i^{\mathrm{lo}} < t \leq d_i$ we have $\tau^{\mathrm{lo}} = 1$, and the third property of Equation 11 gives $\hat{B}(1) \leq \max\{\hat{B}(0), E^{\min}\} \leq E_i(t)$. Lemma S3 with $\tau = 1$ then gives $E_i(t+1) \geq \hat{B}(0) \geq E^{\mathrm{req}}$ again, so $E_i(d_i + 1) \geq E^{\mathrm{req}}$ holds on the event, and the union bound gives the stated probability. If no step of the stay is infeasible almost surely, the second probability vanishes and the bound becomes $1 - \alpha$. The bound does not hold conditionally on the event that A5 held, because that event may be correlated with d_i. □

S2 EV2Gym modelling notes

This section documents how the model of the main text maps onto EV2Gym at commit `55ef368c`, and the simulator is used without changes to its source. All deviations from the defaults of the simulator are made through our own wrapper, configuration files and EV specification file.

S2.1 Charging taper and Assumption A1

In state-of-charge units $\varsigma = E/E^{\mathrm{cap}}$, let $\delta = \bar{P}^{\mathrm{nom}}\Delta t/E^{\mathrm{cap}}$ be the gain of one full-power step, $\varsigma^{\mathrm{tr}} \in (0, 1]$ the transition state of charge, m the taper multiplier and $\kappa = m\delta/(1 - \varsigma^{\mathrm{tr}})$. When the EV's maximum alternating-current charging power does not exceed the charger rating, reading the charging routine of EV2Gym at full current gives

$$\frac{\varphi(E)}{E^{\mathrm{cap}}} = \min\{\hat{\varphi}(\varsigma), \varsigma + \delta\}, \qquad \hat{\varphi}(\varsigma) = \begin{cases} \varsigma + \delta, & \varsigma + \delta \leq \varsigma^{\mathrm{tr}}, \\ 1 - (1 - \varsigma^{\mathrm{tr}}) \exp\Big(- m \dfrac{\varsigma + \delta - \varsigma^{\mathrm{tr}}}{1 - \varsigma^{\mathrm{tr}}} \Big), & \varsigma < \varsigma^{\mathrm{tr}} < \varsigma + \delta, \\ 1 - (1 - \varsigma)\, e^{-\kappa}, & \varsigma \geq \varsigma^{\mathrm{tr}}. \end{cases} \tag{S2}$$

According to a check script released with the code, the closed form agrees with EV2Gym's charging routine to 1.4×10^{-14} kWh. The check covers 768 combinations of EV model, efficiency, transition state of charge, multiplier $m \in \{1, 5\}$ and step length $\Delta t \in \{5, 15\}$ min. [TODO: The authors re-run and confirm this check].

Lemma S6 (A1 in EV2Gym)**.** *Suppose that the maximum alternating-current charging power of the EV does not exceed the charger rating. Then, for every step length, the full-power one-step map of EV2Gym in Equation S2 is continuous, non-decreasing and 1-Lipschitz.*

Proof. At $\varsigma = \varsigma^{\mathrm{tr}} - \delta$ the first two branches of $\hat{\varphi}$ both equal ς^{tr}, and at $\varsigma = \varsigma^{\mathrm{tr}}$ the last two branches both equal $1 - (1 - \varsigma^{\mathrm{tr}})e^{-\kappa}$. The map is therefore continuous, as a minimum of continuous functions. Its branch slopes are 1, $m \exp(\cdot) > 0$ and $e^{-\kappa} \in (0, 1)$, and $\varsigma + \delta$ has slope 1, so the map is non-decreasing as a minimum of non-decreasing functions, and A1 holds. Only the middle branch can have a slope above 1, namely when $m > 1$. On the interval $(\varsigma^{\mathrm{tr}} - \delta, \varsigma^{\mathrm{tr}})$, the function $\hat{h}(\varsigma) = \hat{\varphi}(\varsigma) - \varsigma - \delta$ is strictly concave with $\hat{h}(\varsigma^{\mathrm{tr}} - \delta) = 0$. If $\hat{h}(\varsigma_1) < 0$ at some interior point, concavity gives $\hat{h}'(\varsigma_1) \leq \hat{h}(\varsigma_1)/(\varsigma_1 - \varsigma^{\mathrm{tr}} + \delta) < 0$, so $\hat{\varphi}'(\varsigma_1) < 1$. Wherever the minimum in Equation S2 selects $\hat{\varphi}$ strictly, its slope is thus below 1, and elsewhere the minimum selects $\varsigma + \delta$ with slope 1. The map is therefore piecewise continuously differentiable with slopes in $[0, 1]$ and hence 1-Lipschitz. □

In EV2Gym, $\eta^{\mathrm{c}} = 1$ and $\varphi(E) = E + \Delta t\, \bar{P}(E)$, so a 1-Lipschitz φ is equivalent to a non-increasing $\bar{P}$, the extra condition of Lemma S5. A taper whose power drops faster than $1/(\eta^{\mathrm{c}}\Delta t)$ kW per kWh would violate A1, and certified thresholds would not repair the departure result.

S2.2 Actuation and Assumption A3

The decision variable p_i is the power that EV2Gym reports, bills and sends to the transformer, and it equals the battery-side energy change per step. With this convention $\eta_i^{\mathrm{c}} = \eta_i^{\mathrm{d}} = 1$ in Equation 1, and EV2Gym's charge and discharge efficiencies ζ_i^{c} and ζ_i^{d} act only as derating factors on the power limits.

Algorithm S1 One step of the safety layer

Require: raw action $x \in \mathbb{R}^N$ (non-finite entries set to 0); for each connected EV: $E_i(t)$, $\tau_i(t)$, certified table $\hat{B}_i$; limits $C(t)$, $C_{\mathrm{rev}}(t)$

for each occupied charger i **do**
  $u_i \leftarrow \bar{P}_i(E_i)$; $p_i^{\min} \leftarrow$ Equation 2
  if $E_i \geq \hat{B}_i(\tau_i) - 10^{-9}$ **then** $l_i \leftarrow \min\{u_i, \max\{p_i^{\min}, g_i^{-1}((\hat{B}_i(\tau_i - 1) - E_i)/\Delta t)\}\}$
  else $l_i \leftarrow u_i$
  end if
  $\chi_i \leftarrow (E_i - \hat{B}_i(\tau_i))/(\eta_i^{\mathrm{c}} \bar{P}_i^{\mathrm{nom}} \Delta t)$
end for
$\tilde{C} \leftarrow C - n(t)\varepsilon_q$
if $\sum_i l_i \leq \tilde{C}$ **and** $\sum_i u_i \geq -C_{\mathrm{rev}}$ **and** $\tilde{C} \geq -C_{\mathrm{rev}}$ **then** ▷ feasible step (A5)
  $p^0 \leftarrow \mathrm{clip}(x, l, u)$
  if $\mathbf{1}^\top p^0 \in [-C_{\mathrm{rev}}, \tilde{C}]$ **then** $p \leftarrow p^0$
  else
    sort breakpoints $\{x_i - u_i\} \cup \{x_i - l_i\}$; find λ^* by one pass
    $p \leftarrow \mathrm{clip}(x - \lambda^*, l, u)$; if $\mathbf{1}^\top p \notin [-C_{\mathrm{rev}}, \tilde{C}]$, recompute by safe bisection
  end if
else
  $p \leftarrow$ fallback rule of Section S1.6; log the case
end if
send p to the simulator: full current if $p_i \geq u_i - 10^{-3}$, else rounded upwards on the action grid
return p

Our specification file fixes one efficiency per EV model, 0.90 for seven models and 0.93 for one, taken from the simulator's table at the model's nominal current. EV2Gym rounds the normalised action, the command divided by $\zeta_i^{\mathrm{c}} P^{\mathrm{cs}}$ with the charger rating $P^{\mathrm{cs}} = 22.17$ kW, to five decimals and ignores discharge currents below 0.01 A. It also rounds the stored energy up to 0.01 kWh after every active step without billing the added energy. The actuation map therefore sends a command as full current when it lies within $\varepsilon_{\mathrm{sat}} = 10^{-3}$ kW of u_i, and otherwise rounds it upwards on the five-decimal action grid. Algorithm S1 lists one complete step of the safety layer with this actuation. Rounding can break the aggregate limit when $\sum_i l_i$ equals $\tilde{C}$ to machine precision. The implementation therefore checks the limit after each projection and recomputes a failing row by a bisection that keeps the feasible end of its bracket.

Lemma S7 (A3 in EV2Gym)**.** *Suppose that the maximum alternating-current charging power of every EV does not exceed the charger rating and that efficiencies are constant over current levels. Let chargers and EVs be three-phase, with one EV per charger and zero dead-bands for charging. Assume further that V2G is enabled and that all EVs leave before the episode ends, and let $z_0(m) \in (0, 1)$ be the positive root of $1 - e^{-mz} = z$ for $m > 1$. If $\kappa \leq \kappa^*(m) = m\, z_0(m)$ for every EV, the actuation map realises A3 with $\varepsilon_q = 0.008$ kW.*

Proof. We first show that every sub-maximal charging command $p \in (0, \bar{P}(E)]$, sent as the pilot $p/(\zeta^{\mathrm{c}} P^{\mathrm{cs}})$ of the normalised action, is delivered exactly. Let $\hat{\delta} = p\Delta t / E^{\mathrm{cap}}$ be its gain, with $\hat{\delta} \leq \hat{\delta}^* = \varphi(E)/E^{\mathrm{cap}} - \varsigma \leq \delta$. For a reduced pilot, EV2Gym shifts the transition point to $\varsigma_x = 1 - (\hat{\delta}/\delta)(1 - \varsigma^{\mathrm{tr}}) \geq \varsigma^{\mathrm{tr}}$, and we write $\Omega = 1 - \varsigma_x$. The delivered gain is $\min\{\hat{\delta}, \hat{\varphi}_x(\varsigma) - \varsigma\}$, where $\hat{\varphi}_x$ is the map of Equation S2 with $(\varsigma^{\mathrm{tr}}, \delta)$ replaced by $(\varsigma_x, \hat{\delta})$, and the exponent $m\hat{\delta}/\Omega$ of the shifted taper equals κ.

If $\varsigma + \hat{\delta} \leq \varsigma_x$, the step stays in the constant-current branch and the gain is $\hat{\delta}$. For $\varsigma \geq \varsigma_x$, the taper branch gives $\hat{\varphi}_x(\varsigma) - \varsigma = (1 - \varsigma)(1 - e^{-\kappa})$. Since $\varsigma \geq \varsigma^{\mathrm{tr}}$, Equation S2 gives $\hat{\delta} \leq \hat{\delta}^* \leq (1 - \varsigma)(1 - e^{-\kappa})$, and the gain is again $\hat{\delta}$.

In the remaining case $\varsigma < \varsigma_x < \varsigma + \hat{\delta}$, let $y = \varsigma + \hat{\delta} - \varsigma_x > 0$ and $\psi = y/\Omega$. Then $\hat{\varphi}_x(\varsigma) - \varsigma - \hat{\delta} = \Omega(1 - e^{-m\psi} - \psi)$, a concave function of ψ that vanishes at 0 and at $z_0(m)$, so the gain is $\hat{\delta}$ whenever $\psi \leq z_0(m)$. Since $\psi = 1 - (1 - \varsigma - \hat{\delta})/\Omega$, this condition reads $1 - \varsigma - \hat{\delta} \geq (1 - z_0)\, \hat{\delta}(1 - \varsigma^{\mathrm{tr}})/\delta$. If $\varsigma < \varsigma^{\mathrm{tr}}$, then $1 - \varsigma - \hat{\delta} \geq 1 - \varphi(E)/E^{\mathrm{cap}} \geq (1 - \varsigma^{\mathrm{tr}})e^{-\kappa}$ by Equation S2, and with $\hat{\delta} \leq \delta$ the condition holds if $e^{-\kappa} \geq 1 - z_0 = e^{-mz_0}$, that is, if $\kappa \leq mz_0$. For $\varsigma \geq \varsigma^{\mathrm{tr}}$, the bound $\hat{\delta} \leq (1 - \varsigma)(1 - e^{-\kappa})$ gives $1 - \varsigma - \hat{\delta} \geq (1 - \varsigma)e^{-\kappa}$. The right-hand side of the condition is then at most $(1 - z_0)(1 - \varsigma)(1 - e^{-\kappa})m/\kappa$,

so it suffices that $\kappa/(e^{\kappa}-1) \geq m(1-z_0)$. At $\kappa = mz_0$ the two sides are equal, because $e^{mz_0} = 1/(1-z_0)$, and the left side decreases in κ, so $\kappa \leq mz_0$ suffices again. Discharging has no taper, and EV2Gym delivers a discharge command within $[p^{\min}(E), 0)$ exactly, floored at $E^{\min}$ and capped by the discharge limit.

Next we bound the realised power. A saturated command delivers $\varphi(E) - E = \Delta t\, u_i$, and $u_i - p_i \leq \varepsilon_{\text{sat}} = 10^{-3}$ kW. Any positive command below $u_i - \varepsilon_{\text{sat}}$ is rounded upwards by at most $\zeta^{\text{c}} P^{\text{cs}} 10^{-5} \leq 2.22 \times 10^{-4}$ kW, stays below u_i and is delivered exactly by the first part, and EV2Gym's own rounding leaves a value on the grid unchanged. Negative commands are rounded towards zero by at most 2.22×10^{-4} kW, and if the dead-band applies, the realised power is 0 with $0 - p_i < 0.0069 + 2.22 \times 10^{-4}$ kW. Otherwise $\tilde{p}_i \geq p_i \geq p^{\min}(E)$, so neither the discharge limit nor the energy floor clips the command. In all cases $0 \leq \tilde{p}_i - p_i \leq 0.0072 < \varepsilon_q$ kW, and empty chargers draw zero power.

Finally, the stored energy is $E(t+1) = \lceil E + \Delta t\, \tilde{p}_i \rceil_{0.01} \geq E + \Delta t\, \tilde{p}_i \geq E + \Delta t\, p_i$. Also $E + \Delta t\, \tilde{p}_i \leq \varphi(E) \leq E^{\text{cap}}$, and since every capacity is a multiple of 0.01 kWh that is exactly representable after rounding, the upward rounding cannot exceed E^{cap}. □

With $m = 5$, the root is $z_0(5) = 0.99302$ and the threshold is $\kappa^*(5) = 4.965$. In our configuration, $\kappa \leq 2.391$ at $\Delta t = 15$ min and $\kappa \leq 0.797$ at $\Delta t = 5$ min for all eight EV models, so Lemma S7 applies. For $m \leq 1$ the delivery is not exact near the transition point, and the taper multiplier therefore keeps its default value of 5. Without the tightening of the forward limit by $n(t)\varepsilon_q$, realised powers would exceed the forward limit by up to about $n(t) \cdot 0.007$ kW on tight steps, according to a closed-loop check released with the code.

S2.3 Exogenous headroom (Assumption A6)

EV2Gym scales the peak of the inflexible load to $m_L P_{\text{tr}}$ with $m_L \sim \text{Normal}(0.4, 0.1^2)$ and clips the load at P_{tr}. PV generation is $\text{pv}(t)\, m_P P_{\text{tr}}$ with a normalised profile $\text{pv}(t) \leq 0.853 \cdot 1.1 < 0.94$ and $m_P \sim \text{Normal}(0.4, 0.1^2)$, both drawn once per episode. Hence $C(t) \geq P_{\text{tr}}(1 - m_L)$, and with $P_{\text{tr}} = \rho_{\text{tr}} N$ and $\rho_{\text{tr}} \geq 3$ kW, the condition $C(t) \geq N\varepsilon_q$ can fail only if $m_L > 1 - \varepsilon_q/\rho_{\text{tr}} \geq 0.997$, an event of about six standard deviations. Likewise $C_{\text{rev}}(t) \geq P_{\text{tr}}(1 - 0.94\, m_P)$ is negative only if $m_P > 1.06$, an event of more than six standard deviations. A6 therefore fails with a probability below 2×10^{-9} per episode. Since A6 is exogenous, any violation would show up as a step of case (i), (ii) or (iv) of the fallback rule, so the fallback counts of the result tables check it on every replay.

S2.4 Scenarios, arrival profiles and observation

As seen in Table S2, the grid quantities and the arrival rate scale linearly with the number of chargers N, whereas prices do not. The three arrival profiles of EV2Gym differ in their arrival times, stays and requested energies.

Workplace arrivals concentrate on weekday mornings, public arrivals spread over the day with an afternoon and evening peak, and private arrivals occur in the late afternoon and evening with overnight stays. EV2Gym does not generate an EV whose departure would fall within four steps of the episode end, so overnight sessions are silently missing when the episode starts too late. The public and private profiles therefore start at 12:00 instead of 05:00. On the scenario generator alone, the share of dropped spawn attempts then falls to 6.8% and 3.7%, and spawn multipliers per profile equalise the mean charger occupancy at about 0.21. They also equalise the energy demand per charger to within about 5%, and these scenario-generator statistics are design checks rather than results [TODO: the authors re-measure and confirm them]. The episode starts empty, and loads and PV are redrawn for every episode. Our literature configuration reproduces EV2Gym's 25-charger V2G profit-maximisation case with 100 kW, the workplace profile and no load or PV, using our EV specification file.

Table S3 lists the features of the two token types. All features are 0 for an empty charger, and empty chargers are also masked as attention keys. Forecasts are $\hat{P}(t+k) = \max\{0, P(t+k)(1 + 0.1\epsilon_{t,k})\}$ with independent standard normal $\epsilon_{t,k}$, and beyond the episode end the last true value is held. Every global feature is a ratio that does not grow with N, with $P_{\text{tr}}^{\text{nom}} = \max_t P_{\text{tr}}(t)$ and $N_{\text{day}} = 24/\Delta t$. The arrival profile is not a feature.

Table S2: Scenario parameters. Quantities marked "linear" scale with the number of chargers N.

Item	Setting	Scaling
Simulator	EV2Gym, commit `55ef368c`, unmodified; one EV per charger	–
Station sizes	train $N \in \{10, 20\}$; test $N \in \{5, 10, 20, 50, 100, 200\}$	–
Transformer rating	$P_{\mathrm{tr}} = \rho_{\mathrm{tr}} N$; $\rho_{\mathrm{tr}} \in \{4, 7\}$ kW (tight, moderate); stress level 3 kW at $N \in \{20, 200\}$	linear
Inflexible load	mean peak $0.4P_{\mathrm{tr}}$, clipped at P_{tr}, redrawn per episode	linear
PV generation	mean peak about $0.38P_{\mathrm{tr}}$, redrawn per episode	linear
Demand response	off	–
Step, horizon	$\Delta t = 15$ min, $T = 112$ (28 h); sensitivity 5 min, $T = 336$	–
Profiles (start, spawn multiplier, days)	workplace (05:00, 5, weekdays); public (12:00, 4, all days); private (12:00, 2.5, all days)	per charger
Mean occupancy	about 0.21 for all profiles	–
EV models	8 V2G-capable models; efficiency 0.90 (0.93 for one model); $E^{\min} = 5$ kWh; minimum stay 180 min	–
Taper multiplier	5 (EV2Gym default)	–
Desired energy	full battery; sensitivity 80% state of charge	–
Prices	Dutch day-ahead prices; discharge price = charging price; sensitivities with factor 0.9 and a throughput cost of 0.05 €/kWh	–
Forecast noise	Gaussian, standard deviation 10% of the value	–
Test grid	36 configurations (N, ρ_{tr}, profile), 6 stress configurations and the literature configuration	–
Test episodes	100 per configuration for $N \le 50$ and the literature configuration, 50 for $N \ge 100$; 3,550 per policy	–

Table S3: Features of the charger token ($F_c = 12$) and the global token ($F_g = 38$) with their normalisation.

Charger token ξ_i	Global token ξ_0
1: occupancy flag	1–2: $c^{\mathrm{ch}}(t)$, $c^{\mathrm{dis}}(t)$ $/c^{\mathrm{ref}}$
2–3: $E_i(t)$, E_i^{cap} $/E^{\mathrm{ref}}$	3–10: $c^{\mathrm{ch}}(t+k)$, $k = 1, \dots, 8$, $/c^{\mathrm{ref}}$
4: state of charge E_i/E_i^{cap}	11–12: sine and cosine of the hour of day
5: remaining required energy $(E_i^{\mathrm{req}} - E_i)^+/E^{\mathrm{ref}}$	13: weekend flag; 14: remaining episode fraction $(T-t)/T$
6: remaining steps τ_i (conformal variant: τ_i^{lo}) $/N_{\mathrm{day}}$	15–16: $C(t)$, $C_{\mathrm{rev}}(t)$ $/P_{\mathrm{tr}}^{\mathrm{nom}}$
7: laxity χ_i, clipped to $[-N_{\mathrm{day}}, 2N_{\mathrm{day}}]$, $/N_{\mathrm{day}}$	17–18: measured $P^{\mathrm{inf}}(t)$, $P^{\mathrm{pv}}(t)$ $/P_{\mathrm{tr}}^{\mathrm{nom}}$
8–9: $\bar{P}_i^{\mathrm{nom}}$, P_i^{dis} $/P^{\mathrm{ref}}$	19–34: forecasts of P^{inf} and P^{pv} for $k = 1, \dots, 8$, $/P_{\mathrm{tr}}^{\mathrm{nom}}$
10: physical lower limit $p_i^{\min}(E_i)$ $/P^{\mathrm{ref}}$	35: occupancy ratio $n(t)/N$; 36: rating per charger $P_{\mathrm{tr}}^{\mathrm{nom}}/(NP^{\mathrm{ref}})$
11–12: bounds l_i, u_i $/P^{\mathrm{ref}}$	37–38: $\sum_i l_i$, $\sum_i u_i$ $/P_{\mathrm{tr}}^{\mathrm{nom}}$

S3 Request pipeline and perfect-information optima

S3.1 Station-level request pipeline

The arrival cap and admission control belong to the request setting of the station operator and not to the controller. For every test replay they are computed once, before any controller runs, and the resulting admitted requests E_i^{req} are shared by every compared method and by both optima. The wrapper writes the admitted requirement into EV2Gym's EV object at arrival, so external baselines such as EV-GNN and the Gurobi eMPC pursue the same target. For arrivals inside its horizon, the HiGHS eMPC uses the shared entries. Conformal passes re-certify the shared requirement with their bound $\hat{d}_i^{\mathrm{lo}}$, so their requirement is never larger, and passes without admission control keep the arrival-capped request. In every pass, shortfall is measured against the shared admitted requirement.

Arrivals are processed in time order, with ties broken by the charger index. An arriving EV j enters with its exogenous arrival energy, desired energy, departure step and EV parameters, and the cap of Equation 3 gives its first requirement with a certified threshold table. Admission control then solves the feasibility LP of Equation S3 at the arrival step t_0 over the EVs admitted earlier and still connected, together with EV j:

$$
\begin{aligned}
& E_i(t+1) = E_i(t) + \Delta t\, p_i(t), \qquad E_i^{\min} \le E_i(t+1) \le E_i^{\mathrm{cap}}, \qquad E_i(d_i+1) \ge E_i^{\mathrm{tgt}}, \\
& -P_i^{\mathrm{dis}} \le p_i(t) \le \bar{P}_i^{\mathrm{nom}}, \qquad p_i(t) \le \varrho_i^{\mathrm{in}} \left(E_i^{\mathrm{cap}} - E_i(t)\right) \ (t > t_0), \qquad p_i(t_0) \in [p_i^{\min}, \bar{P}_i], \\
& -\hat{C}_{\mathrm{rev}}(t) \le \textstyle\sum_i p_i(t) \le \hat{C}(t) - n(t_0)\varepsilon_q.
\end{aligned}
\tag{S3}
$$

The energies of the earlier EVs at t_0 are read from a nominal plan and not from the trajectory of any controller. After each admission decision, an elastic version of the same LP updates this plan so that the targets are reached as early as possible. The limits $\hat{C}$ and $\hat{C}_{\mathrm{rev}}$ use the measured values at t_0, the noisy forecasts for the next H_f steps and the last forecast afterwards. Its slope ϱ_i^{in} is the largest value with $\bar{P}_i(E) \geq \min\{\bar{P}_i^{\mathrm{nom}}, \varrho_i^{\mathrm{in}}(E_i^{\mathrm{cap}} - E)\}$ for all E, so the linear taper model is an inner approximation and every LP-feasible schedule is physically realisable.

Inside the LP, the new EV has the target $E_j^{\mathrm{tgt}} = E_j^{\mathrm{req}}$, and an earlier EV has the smaller of E_i^{req} and $\Psi_i^{\mathrm{LP}}(E_i(t_0)) - 10^{-6}$ as its target. Here Ψ_i^{LP} is the largest departure energy reachable in the LP's taper model, and this target is used only inside the admission LP. With full-battery requests as literal targets, the LP would be infeasible whenever such an EV is connected, because the linear taper model never reaches full charge in finitely many steps. Feasibility is monotone in the new request, so one LP that maximises $E_j(d_j + 1)$ gives the largest admissible request. Requests admitted earlier are never reduced, and the reduced request of EV j passes through the cap routine again, so A4 holds for every EV and every method. If the LP is infeasible even for the lowest request, the request is set to the arrival energy.

Each decision uses only information available at t_0, so a station operator can run the pipeline online. If its LP is feasible, an open-loop schedule exists that meets all targets from the nominal-plan energies under the forecast limits, provided that no further EV arrives. The pipeline does not guarantee A5 for any controller, because realised energies may lie below the nominal plan, forecasts are noisy and later arrivals add load. It therefore does not enter Theorem 1, and the evaluation passes without admission control measure its effect.

S3.2 Linear programs of the optima

Both perfect-information LPs know all sessions of the episode with their arrival and departure steps, arrival energies and admitted requirements, together with all prices and the realised limits $C(t)$ and $C_{\mathrm{rev}}(t)$. For every session i and step $t \in \mathcal{T}_i = \{t_i^{\mathrm{arr}}, \dots, d_i\}$ the variables are $p_i^+(t), p_i^-(t) \geq 0$ and $E_i(t)$, together with a shortfall slack $z_i \geq 0$ and, in the soft LP, overload slacks $o^{\pm}(t) \geq 0$. With the rounding allowance $G_i^{\max} = 0.01\,(d_i - t_i^{\mathrm{arr}} + 1)$ kWh, the constraints are

$$
\begin{aligned}
&E_i(t_i^{\mathrm{arr}}) = E_i^{\mathrm{arr}}, \qquad E_i(t+1) = E_i(t) + \Delta t\,\big(p_i^+(t) - p_i^-(t)\big), \qquad E_i^{\min} - G_i^{\max} \leq E_i(t+1) \leq E_i^{\mathrm{cap}},\\
&p_i^+(t) \leq \bar{P}_i^{\mathrm{nom}}, \qquad p_i^-(t) \leq P_i^{\mathrm{dis}}, \qquad p_i^+(t_i^{\mathrm{arr}}) \leq \bar{P}_i(E_i^{\mathrm{arr}}), \qquad p_i^+(t) \leq \varrho_i^{\mathrm{out}}\big(E_i^{\mathrm{cap}} - E_i(t)\big)\ (t > t_i^{\mathrm{arr}}),\\
&E_i(d_i+1) + z_i \geq E_i^{\mathrm{req}} - G_i^{\max}, \qquad -C_{\mathrm{rev}}(t) - o^-(t) \leq \textstyle\sum_i \big(p_i^+(t) - p_i^-(t)\big) \leq C(t) + o^+(t),
\end{aligned}
\tag{S4}
$$

where the hard LP fixes $o^{\pm} = 0$. The profit of a schedule is $\Pr(p) = \sum_t \Delta t \sum_i \big(c^{\mathrm{dis}} p_i^- - c^{\mathrm{ch}} p_i^+ - c^{\mathrm{thr}}(p_i^+ + p_i^-)\big)$, with the throughput cost c^{thr} equal to zero except in one sensitivity study.

Lemma S8 (Taper rows)**.** *Let every EV satisfy the conditions of Lemma S7, and let $\varrho_{\kappa,i} = (1 - e^{-\kappa_i})/\Delta t$. Then $\bar{P}_i(E) \leq \min\{\bar{P}_i^{\mathrm{nom}}, \varrho_{\kappa,i}(E_i^{\mathrm{cap}} - E)\}$ for every $E \in [E^{\min}, E^{\mathrm{cap}}]$. Equality holds except on the one-step transition band $\varsigma \in (\varsigma_i^{\mathrm{tr}} - \delta_i, \varsigma_i^{\mathrm{tr}})$, and $\sup_E \bar{P}_i(E)/(E_i^{\mathrm{cap}} - E) = \varrho_{\kappa,i}$.*

Proof. In state-of-charge units, $\bar{P}\Delta t/E^{\mathrm{cap}} = \varphi/E^{\mathrm{cap}} - \varsigma$ by Equation S2, and for $\varsigma \geq \varsigma^{\mathrm{tr}}$ this gain is $\min\{\delta, (1-\varsigma)(1-e^{-\kappa})\}$ and equals the right-hand side. The constant-current region $\varsigma + \delta \leq \varsigma^{\mathrm{tr}}$ has the gain δ, and the right-hand side equals δ if and only if $\delta \leq (1-\varsigma)(1-e^{-\kappa})$. Since $1 - \varsigma \geq 1 - \varsigma^{\mathrm{tr}} = m\delta/\kappa$, it suffices that $\kappa \leq m(1 - e^{-\kappa})$, and with $\kappa = mz$ this is $z \leq 1 - e^{-mz}$, that is, $\kappa \leq \kappa^*(m)$. On the transition band, let $y = \varsigma^{\mathrm{tr}} - \varsigma \in (0, \delta)$, so that $1 - \varphi/E^{\mathrm{cap}} \geq (1 - \varsigma^{\mathrm{tr}})e^{-m(\delta - y)/(1-\varsigma^{\mathrm{tr}})}$. With $e^v \geq 1 + v$ and $m \geq 1$, the right-hand side of this bound is at least $e^{-\kappa}(1 - \varsigma^{\mathrm{tr}} + my) \geq e^{-\kappa}(1 - \varsigma)$. The gain is therefore at most $(1-\varsigma)(1-e^{-\kappa})$, and it is at most δ anyway, so the inequality holds everywhere, and the supremum follows from equality in the upper region. □

The code uses $\varrho_i^{\mathrm{out}} = 1.01\,\varrho_{\kappa,i}$, a valid outer row that is 1% looser than exact. A check script released with the code confirms the inequality to 1.7×10^{-13} kW and the equality off the band on 320 configurations. With these rows the LPs model EV2Gym's charging limit up to the step discretisation of the transition band and the 1% margin. The remaining relaxations are the rounding allowance and simultaneous charging and discharging, and the latter is cost- and energy-neutral, and hence exact, when $\eta = 1$, $c^{\mathrm{dis}} = c^{\mathrm{ch}}$ and $c^{\mathrm{thr}} = 0$, as in the main grid. In the sensitivities with a throughput cost or a discharge

price factor of 0.9 at negative prices, the LPs remain valid upper bounds. They may then use simultaneous charging and discharging, and the number of such steps is reported.

S3.3 Upper bounds

The hard LP is solved lexicographically with HiGHS [4]. Stage 1 computes the least total shortfall $z^* = \min \sum_i z_i$ with $G^{\max} = 0$, and stage 2 maximises $\Pr(p)$ with the allowance and $\sum_i z_i \le z^* + 10^{-6}$, giving $J^{\mathrm{LP,ceil}}$. The same stage 2 with $G^{\max} = 0$ gives J^{LP}, reported only in Section S5. In contrast, the soft LP maximises $\Pr(p) - w_E \sum_i z_i - w_P \Delta t \sum_t (o^+(t) + o^-(t))$ subject to Equation S4 in a single stage, giving $J^{\mathrm{pen,LP}}$.

Proposition S2 (Hard optimum)**.** *Let the configuration of Lemma S7 hold, and let π be any controller whose reported powers satisfy Equation 5 at every step and whose realised total shortfall against the admitted requirements is at most z^*. Then $J^\pi \le J^{\mathrm{LP,ceil}}$.*

Proof. Let $\tilde{p}_i(t)$ be the reported powers and $E_i^{\mathrm{nom}}(t) = E_i^{\mathrm{arr}} + \Delta t \sum_{s<t} \tilde{p}_i(s)$ the energy that the billed powers alone would produce. EV2Gym bills exactly $\tilde{p}$, so J^π equals Pr at $p^\pm = \tilde{p}^\pm$. The rounding adds less than 0.01 kWh per active step, so $E_i^{\mathrm{real}}(t) - G_i^{\max} \le E_i^{\mathrm{nom}}(t) \le E_i^{\mathrm{real}}(t) \le E_i^{\mathrm{cap}}$ along the trajectory. Both the lowered energy floor and the lowered departure row therefore hold for E^{nom} with $z_i = (E_i^{\mathrm{req}} - E_i^{\mathrm{real}}(d_i + 1))^+$, and the power rows hold because $\tilde{p}_i \le \bar{P}_i(E_i^{\mathrm{real}}) \le \bar{P}_i^{\mathrm{nom}}$ and $\tilde{p}_i \ge -P_i^{\mathrm{dis}}$. Moreover, the first-step row holds because $E^{\mathrm{nom}}(t_i^{\mathrm{arr}}) = E_i^{\mathrm{arr}}$. For the taper rows, EV2Gym never delivers more than $\varphi(E) - E$, so Lemma S8 gives $\tilde{p}_i(t) \le \varrho_{\kappa,i}(E^{\mathrm{cap}} - E_i^{\mathrm{real}}(t)) \le \varrho_i^{\mathrm{out}}(E^{\mathrm{cap}} - E_i^{\mathrm{nom}}(t))$. The transformer rows are the assumption, and $\sum_i z_i \le z^*$, so the trajectory is feasible for stage 2 and its objective equals J^π. □

Proposition S3 (Soft-penalty optimum)**.** *Let the configuration of Lemma S7 hold, and let commands be clipped to the physical limits of Equation 2. For every simulated trajectory on a replay, with shortfall measured against the shared admitted requirement, $J^{\mathrm{pen}}(\pi) \le J^{\mathrm{pen,LP}}$.*

Proof. Take $p^\pm = \tilde{p}^\pm$, $E = E^{\mathrm{nom}}$, $z_i = \Delta E_i^{\mathrm{sh}}$ and $o^\pm(t)$ equal to the realised overloads in either direction. The energy, power, first-step and taper rows hold as in the proof of Proposition S2, because they use only the physics of EV2Gym and not any constraint that the controller respects. For the departure row, $E^{\mathrm{nom}}(d_i + 1) + z_i \ge E^{\mathrm{real}}(d_i + 1) - G_i^{\max} + z_i \ge E_i^{\mathrm{req}} - G_i^{\max}$, and the transformer rows hold by the definition of $o^\pm$. At this point the objective equals $J^{\mathrm{pen}}(\pi)$, with the same billed powers, shortfall and overload energy. □

The bound is a property of the measurement, not of the controller, so it covers controllers with shortfall or overload, the conformal and no-admission passes and the external baselines. Every pass in the wrapper applies the physical clip, and the Gurobi eMPC adapter clips its commands in the same way. A unit test released with the code checks $J^{\mathrm{pen}} \le J^{\mathrm{pen,LP}}$ for all rule-based methods on small replays, including random actions. Inside EV2Gym, a controller can exceed J^{LP} through the unbilled rounding gain, so the main text uses $J^{\mathrm{LP,ceil}}$ for the cost of the guarantee and reports J^{LP} only here.

S4 Compared methods, training and statistics

S4.1 Implementation of the compared methods

Every method and every evaluation pass receives the shared admitted requirement of Section S3, and shortfall is always measured against it. Methods without the safety layer execute only the physical clip $\mathrm{clip}(x_i, p_i^{\min}, u_i)$, and variants with the layer pass their output as the raw action x through Algorithm S1, so they inherit Theorem 1.

Penalty-based methods. Attention+penalty, MLP+penalty and the variant without departure bound learn from the reward $r(t)$ minus two penalties. They are w times the departure shortfall of the EVs that leave at step t and w times the transformer overload energy of the step. Shortfall is measured against the admitted requirement, overload uses the untightened limits and the reported powers, and the weight

$w \in \{1, 10, 100\}$ €/kWh is tuned. A dense ablation adds $w\Delta t \sum_i (l_i - p_i)^+$ at every step and thus gives the penalty method the information of our lower bound. At the evaluation weight of 10 €/kWh, more than eleven times the highest hourly price, no violation pays at the optimum of the penalised objective, although learned policies can still violate.

Attention+penalty and MLP+penalty use $\beta = 0$, and their actor loss masks only tokens whose action cannot influence the commanded power. The variant without departure bound runs the safety layer with $l_i = p_i^{\min}$, so it keeps $\beta = 0.01$ and masks fallback steps like SA-Attn. For its dense ablation, Attention+penalty keeps the tuned weight w and the curriculum ablation of SA-Attn trains only with $N = 10$ during the first 20% of the steps.

Attention+PPO-Lagrangian. This baseline solves the constrained problem $\max J_r$ subject to expected episode shortfall and overload energy of at most zero [5,6]. The attention policy runs with the physical clip and the profit reward, and the critic has three heads, one for the value and one for each cost, each with its own advantage estimate. PPO uses the combined advantage $(A_r - \Lambda_E A_E - \Lambda_P A_P)/(1 + \Lambda_E + \Lambda_P)$, normalised per minibatch. Once per rollout, $\Lambda_k \leftarrow \mathrm{clip}(\Lambda_k + \eta_\Lambda(\hat{J}_k - 0), 0, 1000)$ for $k \in \{E, P\}$, with $\Lambda_k = 0$ at the start and the step size $\eta_\Lambda \in \{0.5, 2, 8\}$ tuned. The cost limit of zero matches the hard requirements of the problem, and the cap of 1000 keeps the multipliers finite while violations persist, and the training logs record their trajectories.

Size-agnostic and fixed-size networks. Deep Sets+safety layer embeds each charger token with a shared network and adds the masked mean of all embeddings and the global token as context. It decodes one Gaussian per token, so no token attends to another. The MLP baselines map the zero-padded concatenation of 20 charger tokens and the global token through two hidden layers of 640 units. The width matches the parameter count of the attention actor (614,440 versus 610,370), and the MLP baselines are evaluated and reported only at $N \leq 20$.

Rules. The rules as fast as possible (AFAP), as late as possible (ALAP), round robin and LLF are price-unaware and charge only. For each EV, sLLF [7] computes the next-step laxity $\chi_i^0 = (E_i - \hat{B}_i(\tau_i - 1))/(\bar{P}_i^{\mathrm{nom}}\Delta t)$ and the rate $p_i = \mathrm{clip}(\bar{P}_i^{\mathrm{nom}}(L - \chi_i^0), 0, u_i)$, with a common level L found by bisection so that $\sum_i p_i = \max(0, \tilde{C})$. It is the published rule with the peak rate $\bar{P}_i^{\mathrm{nom}}$ and a cap that includes the taper. The price-threshold rule compares the current price with the median of all prices seen so far together with the known price window. Below the median it allocates $\tilde{C}(t)$ by LLF, above the median it requests the maximum V2G discharge, and otherwise it idles.

Model-predictive baselines. eMPC-HiGHS is our re-implementation of the eMPC of EV2Gym [8] as a receding-horizon net-power LP with HiGHS. Its LP uses per-EV limits, exact first-step bounds including the taper and the admitted requirements with an elastic departure constraint of 1000 €/kWh. Prices and arrivals are known exactly within the horizon of 10 steps (eMPC-HiGHS) or 96 steps (eMPC-HiGHS-24h), whereas load and PV come from EV2Gym's noisy forecasts. The original eMPC (Gurobi) runs only on the literature configuration, through an adapter subclass that fixes two incompatibilities with newer numerical libraries and clips its commands to the physical limits.

External learned baselines. EV-GNN [9] runs unmodified at commit `01aa1b7a` with its own twin delayed deep deterministic policy gradient (TD3) script, reward and published hyperparameters, through a launcher that restores numerical-library aliases. It trains on its own EV2Gym configuration for each profile at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW, and this configuration keeps one load and PV draw for the whole training run. Its budget is 3×10^6 steps instead of its default of 10^7, and the checkpoint with the best training reward is evaluated. Both differences to our own methods are stated with the results. For EV-GNN+safety layer, its bounded per-charger action a_i^{GNN} is mapped to the raw action $x_i = a_i^{\mathrm{GNN}} \zeta_i P^{\mathrm{cs}}$. The graph-enhanced decision transformer (GNN-DT) [10] has public code under an open-source licence (DT4EVs, commit `8b327d12`). GNN-DT trains offline on trajectories that an optimiser generates for each scenario inside a modified simulator copy. Porting this pipeline to our scenarios, estimated at two to four person-days, was not done for this study.

S4.2 Training loop

Algorithm S2 lists the training loop of SA-Attn, in which the safety layer of Algorithm S1 acts as part of the environment. Each environment writes into its own buffer, and at update time all token tensors are right-padded to the largest N in the batch with masked entries. Padding changes neither the actor

Algorithm S2 PPO training of SA-Attn with a variable number of chargers

Require: 16 environments (8 with $N = 10$, 8 with $N = 20$); actor π_θ and critic V_ω

while fewer than 3×10^6 environment steps **do**
  for 128 steps in every environment **do**
    observe tokens, l, u, $\tilde{C}$, C_{rev} and the feasibility flag
    sample $a \sim \pi_\theta(\cdot \mid s)$; $x \leftarrow P^{\text{ref}} a$
    $p \leftarrow \Pi_t(x)$ if feasible, else fallback rule ▷ Algorithm S1
    step the simulator; $r \leftarrow$ realised profit of the step (Equation 6); store $r/(Nr^{\text{ref}})$, a, $\log \pi_\theta(a \mid s)$ over tokens with $u_i - l_i > 10^{-9}$
  end for
  compute advantages with GAE ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$); pad tokens to the largest N in the batch
  for 10 epochs, minibatches of n_{mb} transitions **do**
    clipped surrogate ($\epsilon_{\text{clip}} = 0.2$) on feasible transitions with normalised advantages, divided by n_{mb}
    loss $\leftarrow -$surrogate $+ c_{\text{vf}}\cdot$value loss $- c_{\text{ent}}\cdot$entropy $+ \mathcal{L}_{\text{reg}}$ (Equation 13, feasible steps only)
    Adam step with gradient-norm clipping 0.5; stop the epochs early if the approximate KL divergence exceeds 0.03
  end for
end while

outputs of real tokens nor the value (Lemma S1), and the log-probability, entropy and regulariser sums run over masks only. The learning reward is the profit divided by Nr^{ref}, so that returns of stations with different N have the same scale, and every reported result uses the undivided profit.

S4.3 Hyperparameters and tuning

Table S4 lists the hyperparameters. Each of the seven tuned learned methods (SA-Attn, Attention+penalty, Attention+PPO-Lagrangian, Attention+safety layer without departure bound, Deep Sets+safety layer, MLP+safety layer and MLP+penalty) receives 12 configurations. They consist of the default point and 11 points sampled without replacement from the grid of Table S4 with a fixed sampling seed per method. Each configuration is trained on the training mixture for 0.5×10^6 steps with seed 0 and evaluated on 24 validation episodes that cycle through the training configurations, with seeds disjoint from the test replays. Selection uses the mean validation J^{pen} at $w = 10\,€/\text{kWh}$ for every method. The conformal variant, the ablations of SA-Attn, the 5-min policy and the reference trained at $N = 50$ inherit the selection of SA-Attn, and the penalty ablations inherit that of Attention+penalty. EV-GNN keeps its published hyperparameters, and β and the architecture are fixed for all methods with the safety layer.

S4.4 Statistical analysis

The training seed is the unit of analysis, and replay-level values are averaged within each seed first. Seed counts are fixed in advance at five, three for the reference trained at $N = 50$ and for $w \in \{1, 100\}$, and there is no adaptive seed rule. Configurations are strata, so pooled statistics average over configurations and bootstrap resampling stays within each configuration. The four pre-declared primary endpoints are E1, the gap at $N = 200$; E2, the cost of the guarantee per N; E3, the overload energy; and E4, the share of EVs without infeasible steps. Point estimates come with t-based and stratified percentile-bootstrap 95% confidence intervals over the seed-level aggregates.

Comparisons with SA-Attn run per station size on the per-table metrics, and each safety-layer method is also compared with its penalty-based twin. The primary test is a paired t-test on seed means, pairing training seed k of the two methods on the same fixed replays. Against a deterministic baseline, a one-sample t-test is applied to the seed-level differences, and Welch's test is used when seed indices do not match. The unpaired Welch test on the same seed means is reported as a robustness check. As a secondary test, Wilcoxon signed-rank tests [11] compare the replay-level differences of the seed-averaged policies. The Holm correction runs over all comparisons of one test file, and each file covers one results table. Both the interquartile mean of the gap and the probability of improvement are reported with

Table S4: Hyperparameters and tuning grid.

Group	Parameter	Value
Network	width d_m, layers n_L, heads n_H	128, 3 (pre-LayerNorm), 4
	feed-forward width, dropout	512 (GELU), 0
	actor head (shared), critic readout	$128 \to 64 \to 2$; attention pooling with one query, $128 \to 64 \to 1$
	$\log \sigma$ clamp and initial value	$[-5, 0.5]$, -0.5
	parameters (actor / critic)	610,370 / 676,481; small network 228,067 in total (head width 32)
	precision	float32 networks, float64 safety layer
PPO	optimiser	Adam, $\epsilon = 10^{-5}$, linear learning-rate decay
	learning rate (tuned)	$\{1, 3, 10\} \times 10^{-4}$
	minibatch n_{mb} (tuned)	$\{256, 512, 1024\}$
	entropy coefficient c_{ent} (tuned)	$\{0, 10^{-3}, 10^{-2}\}$
	discount γ, GAE λ_{GAE}	0.99 ($0.99^{1/3}$ at 5 min), 0.95
	clip range ϵ_{clip}, value coefficient c_{vf}	0.2, 0.5
	environments, rollout length	16 (8 with $N = 10$, 8 with $N = 20$), 128
	epochs, target KL, gradient-norm clip	10, 0.03, 0.5
	advantage normalisation	per minibatch
	steps per run, seeds	3×10^6; 5 (3 for the $N = 50$ reference and $w \in \{1, 100\}$)
Penalty reward	training weight w (tuned), β	$\{1, 10, 100\}$ €/kWh; $\beta = 0$, except 0.01 without departure bound
Lagrangian	dual step η_Λ (tuned), cost limit, $\Lambda_{\max}$	$\{0.5, 2, 8\}$, 0, 1000
Safety layer	P^{ref}, β	11 kW, 0.01
	ε_q, $\varepsilon_{\mathrm{sat}}$, tolerance	0.008 kW, 0.001 kW, 10^{-9} kW or kWh
Features	E^{ref}, c^{ref}, H_f	100 kWh, 0.25 €/kWh, 8 steps
Evaluation	weights w_E, w_P in J^{pen}	10 €/kWh each, all methods
	test action	deterministic, $a = \mu_\theta$
Conformal	n_{cal}, α	5,000 per configuration, $\{0.01, 0.05, 0.1, 0.2\}$
	quantile predictor	gradient boosting, 200 trees of depth 3, 2,000 training sessions

stratified-bootstrap intervals [12]. Henderson et al. showed that results over few seeds can mislead [13], and with five seeds the tests have low power, so the confidence intervals are the primary evidence.

S4.5 Compute budget

The core set needs the tuning runs of seven methods, five seeds of eight trained methods and three seeds of the reference trained at $N = 50$. It also contains the admission tables, the conformal calibrations, about 89 evaluation passes of 3,550 episodes each and the optima of all replays. Together, the core items need about 32 GPU-hours of training and 82–132 CPU-hours for evaluation, admission tables, calibration and optima. The extended set adds the ablations, the penalty-weight study, the sensitivity studies, eMPC-HiGHS-24h, the Gurobi eMPC on the literature configuration and EV-GNN, for about 31–41 further GPU-hours and 68–108 CPU-hours. Without retraining, the conformal variant is evaluated at all four values of α, because Proposition 2 holds for any policy. [RESULT: measured training time per run on the hardware of the main text, Table S11].

S5 Additional results

[TODO: Skeleton: fill the result placeholders from the analysis outputs, use the past tense and write the interpretation yourself].

The tables of this section are generated by the analysis pipeline. Its files T2, T3, T3b, T5, T4, T7, T6, TS1, TS2 and TS3 appear as Tables S5 to S14 in this order.

S5.1 Learning curves, transfer, constraints and cost of the guarantee

Figure S1 shows the training curves of the learned methods. Across seeds, the training curves of SA-Attn [RESULT: description of convergence and seed spread, Figure S1]. Table S5 lists the zero-shot results

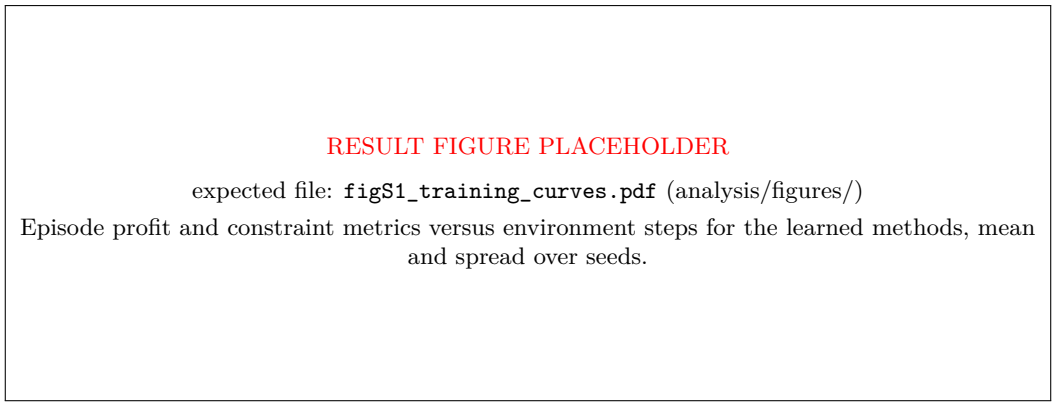


Figure S1: Training curves of the learned methods.

Table S5: Zero-shot transfer across station sizes N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

behind Figure 3 for every station size, including the number of fallback steps per episode and the transfer gap to the reference trained at $N = 50$. That reference is in distribution only at $N = 50$, so at $N = 100$ and $N = 200$ the transfer gap compares two extrapolations. Pooled over the test grid, Table S6 gives E4, energy not served, overload, the shortfall relative to the deficit bound of Equation S1 and the capped and admission-reduced requests. The deficit-bound ratio covers only the methods with the full safety layer, because the variant without departure bound has no such bound.

Per station size, Table S7 gives E4, the infeasible steps by fallback case and the binding rates of the aggregate limits and of the departure lower bound. Table S8 gives the cost of the guarantee behind Figure 5, in J^{pen} and in profit, with the hard-optimum gaps of both twins. The hard-optimum gap of a method with violations is not bounded below by zero, because Proposition S2 does not cover it.

S5.2 Ablation study and further baselines

Figure S2 and Table S9 compare attention and MLP policies with and without the safety layer, the further baselines and the ablations of SA-Attn. The variants without departure bound, with the safety layer at deployment only and without admission control give [RESULT: gaps relative to SA-Attn, Table S9]. Setting $\beta = 0$, training with a curriculum and using the small network give [RESULT: gaps relative to SA-Attn, Table S9], and the dense penalty gives [RESULT: gap of Attention+penalty with the dense penalty, Table S9].

S5.3 Conformal departure bound

Figure S3 and Table S10 show coverage, service and profit of SA-Attn-CP against the miscoverage level α. The conformal variants re-certify the shared requirement with their bound, and their shortfall is measured against the shared requirement, so the table also reports their main gap. Averaged over episodes, the coverage is [RESULT: coverage for each α against the nominal level $1 - \alpha$, Table S10]. Compared with known departure steps, the profit changes by [RESULT: profit difference for each α, Table S10], and the uncalibrated point prediction gives [RESULT: coverage and energy not served of the point prediction, Table S10].

S5.4 Computation time

Figure S4 and Table S11 report the inference time per control step and the training time per run. At $N = 10$ and $N = 200$, one control step of SA-Attn takes [RESULT: mean inference time in ms per step, Table S11], and one training run takes [RESULT: training time per run in hours, Table S11].

Table S6: Constraint satisfaction per method over all test-grid configurations, including E4 (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S7: E4, infeasible steps by fallback case and binding rates per station size N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

S5.5 Hard optima, sensitivity studies and literature configuration

For every station size, Table S12 reports the profit gap of the main methods with respect to $J^{\mathrm{LP,ceil}}$ and to J^{LP}. The two gaps of SA-Attn are [RESULT: both gaps per N, Table S12]. Separately, Table S13 covers the main methods in the sensitivity studies at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW, the stress level $\rho_{\mathrm{tr}} = 3$ kW and the penalty weights 1 and 100 €/kWh. A desired state of charge of 0.8 gives [RESULT: gap and energy not served, Table S13], and a step of 5 min gives [RESULT: gap and energy not served, Table S13]. With the throughput cost and the discharge price factor, the results are [RESULT: gap and profit, Table S13]. On EV2Gym's literature configuration, without the MLP baselines, Table S14 lists [RESULT: gap, profit and energy not served per method, Table S14].

S5.6 Pipeline, pooled constraints and example episode

Figure S5 summarises the experimental pipeline. Pooled over station sizes, the constraint metrics of every method appear in Figure S6. Finally, Figure S7 traces one test episode of SA-Attn with the aggregate power against the forward and reverse limits, the band between $\sum_i l_i$ and $\sum_i u_i$, and the price. In this episode, SA-Attn [RESULT: description of how charging and V2G discharging follow the price and the limits, Figure S7].

Table S8: Cost of the guarantee per station size N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

RESULT FIGURE PLACEHOLDER

expected file: `figS2_ablation.pdf` (analysis/figures/)

Optimality gap and constraint metrics of the ablation variants.

Figure S2: Ablation study.

Table S9: Ablation of architecture, constraint handling and training choices, with the further baselines (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

RESULT FIGURE PLACEHOLDER

expected file: `figS3_conformal_alpha.pdf` (analysis/figures/)

Coverage, share of unserved EVs and profit change versus α.

Figure S3: Trade-off of the conformal departure bound.

Table S10: Conformal departure bound: miscoverage level α versus gap, coverage, profit and energy not served (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

RESULT FIGURE PLACEHOLDER

expected file: `figS4_inference_time.pdf` (analysis/figures/)

Inference time per control step versus N.

Figure S4: Inference time per control step.

Table S11: Computation: training time per run and inference time per control step (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S12: Profit gap with respect to $J^{\mathrm{LP,ceil}}$ and J^{LP} per station size N (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S13: Sensitivity studies at $N = 20$ and $\rho_{\mathrm{tr}} = 4$ kW, stress level and penalty weights (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

Table S14: Results on EV2Gym's 25-charger literature configuration (placeholder until the analysis scripts have run).

[RESULT: table generated by `analysis/scripts/make_all.py`]

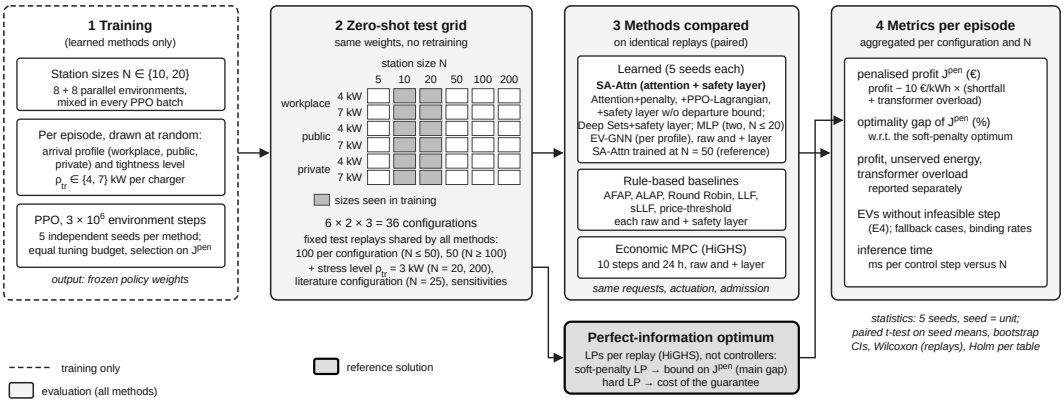


Figure S5: Experimental pipeline. Learned methods train on mixed station sizes, arrival profiles and ratings, and all methods run on identical test replays with the same admitted requirements. The main gap uses the soft-penalty optimum, and the cost of the guarantee uses the hard optimum.

RESULT FIGURE PLACEHOLDER

expected file: `figS6_constraints_pooled.pdf` (analysis/figures/)

Energy not served and overload energy per method and transformer rating, pooled over station sizes.

Figure S6: Constraint metrics per method, pooled over station sizes.

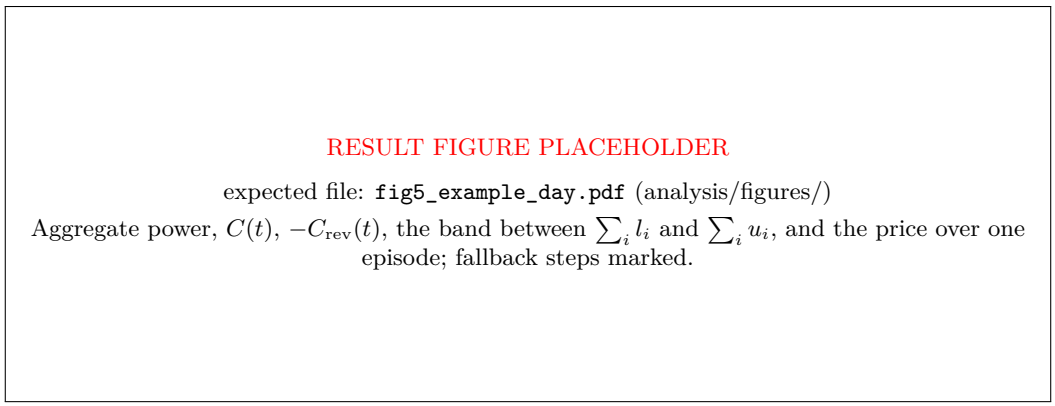


Figure S7: Example episode of SA-Attn (workplace profile, $N = 20$, tight rating).

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